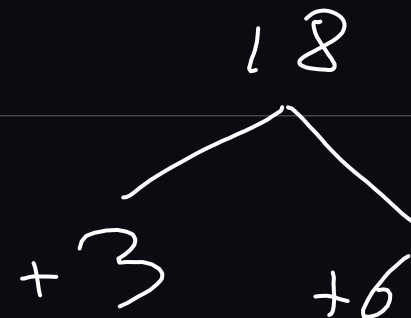


6.1 Q.E

$$x^2 + 9x + 18 = 0$$

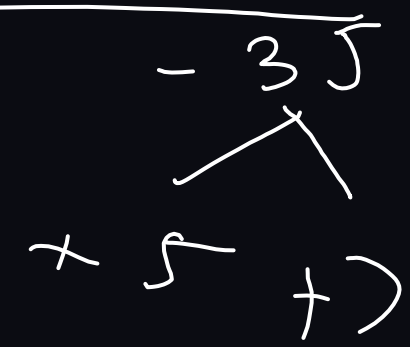


$$x^2 + 3x + 6x + 18$$

$$x(x+3) + 6(x+3)$$

$$(x+3)(x+6)$$

$$p^2 + 2p - 35$$



$$\underline{(a+b)^2 = a^2 + 2ab + b^2}$$

$$(a+b)(a-b) = a^2 - b^2$$

$$\underbrace{(x+y)^3}_a - \underbrace{(x-y)^3}_b$$

$$(a-b)(a^2 + ab + b^2)$$

$$[(x+y) - (x-y)] [(x+y)^2 + (x+y)(x-y) + (x-y)^2]$$

$$[\cancel{x+y} - \cancel{x+y}] [(\underline{x^2} + \cancel{2xy} + \underline{y^2}) + \underline{x^2} - \underline{y^2} + \underline{x^2} - \cancel{2xy} + \underline{y^2}]$$

$$2y \times (3x^2 + y^2)$$

$$6x^2y + 2y^3$$

$$\boxed{y' \times y^2}$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$x^2 + 8x - 48 = 0$$

$$x^2 + 8x + 16 - 16 - 48 \div 2$$

$$(x+4)^2 - 64 = 0 \quad (4)^2 = 16$$

$$(x+4)^2 = 64$$

$$(x+4)^2 = (\pm 8)^2$$

$$x+4 = 8 \quad \text{OR} \quad x+4 = -8$$

$$x = 8 - 4 \quad x = -8 - 4$$

$$x = 4$$

$$x = -12$$

$$x^2 + 8x - 48$$

$$x^2 + 8x - 48$$

$$x^2 + 12x - 4x - 48$$

$$x(x+12) - 4(x+12)$$

$$(x+12)(x-4)$$

$$\begin{array}{c} 48 \\ \swarrow \searrow \\ 12 \quad 4 \end{array}$$

$$24a^3 + 81b^3$$

$$3(8a^3 + 27b^3)$$

$$2\pi\gamma = 22$$

$$\gamma = \frac{22}{2\pi}$$

$$x+1 = \sqrt{6} \quad \text{OR } x+1 = -\sqrt{6}$$

$$x = \sqrt{6} - 1$$

$$x = -1 - \sqrt{6}$$